

Examining the Relational Content and Personal Determinants of Social Signatures Across Multiple Time Scales

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Abstract

How do individuals allocate cognitive and emotional bandwidth across personal networks? Using smartphone logs pooling all outgoing communication channels (calls, SMS, MMS, and WhatsApp) and eight waves of ego-network surveys from the *NetHealth* study ($N = 515$ egos, 15.6 million events), this paper examines the structure, stability, and psychological foundations of social signatures, the ranked distribution of communication effort directed to alters. Intraindividual self-divergence is consistently smaller than interindividual reference divergence ($p < 0.0001$). Power-law decay models fit well (mean $R^2 = 0.933$), though an exponential alternative becomes competitive once texting is pooled with calls (power-law preferred in 61% of signatures). Linking communication ranks to ego-network surveys ($N = 21,273$ dyads) shows the primary signature slot is friend-dominated rather than kinship-dominated, with emotional support and advice declining monotonically across ranks while companionship remains elevated throughout. Stability persists despite substantial alter turnover (73.5% per semester) and despite the top-volume alter changing identity in most semesters, indicating a slot-filling mechanism. Multilevel models show turnover and volume shifts, not network clustering, drive temporal change in divergence. Negative Emotionality predicts steeper, more concentrated signatures ($\beta = 0.076, p = 0.0016$), though Big Five traits explain modest variance ($R^2 = 0.058$) once channels are pooled.

Keywords: Social Signatures, Egocentric Networks, Jensen-Shannon Divergence, Tie Strength, Slot-Filling Hypothesis, Personality Traits.

1 Introduction

A central puzzle in the scientific study of human sociality is the tension between relational fluidity and structural constraint [1, 2]. Across the life course, individuals move through dynamic, evolving social worlds: entering new institutions, forming romantic partnerships, making new friends, and allowing older ties to lapse [3]. Yet despite the continual turnover of interaction partners, personal communication behavior is not randomly distributed across contacts, nor does it restructure arbitrarily when network composition changes. Instead, relational investment is bounded by deep-seated cognitive and temporal constraints on sociality [4–6]. While individuals can recognize hundreds of faces and maintain casual awareness of extensive social circles, active personal networks are structured into concentric, hierarchical layers: an intimate support clique of 1 to 2 confidants, a sympathy group of approximately 5 close ties, an affinity group of 12 to 15 regular contacts, and expanding outer bands of declining emotional intensity and contact frequency [7–9].

In a seminal contribution, Saramäki et al. [10] characterized this relational hierarchy using the concept of *social signatures*, namely, the probability distribution of communication effort that an individual (ego) directs toward their contacts (alters), ordered from most to least frequently contacted within a given observation window. Tracking young adults making the transition from high school to university using mobile call detail records, Saramäki et al. [10] uncovered two fundamental empirical regularities: first, social signatures exhibit marked *interindividual variation*, ranging from steep, hyper-concentrated profiles focused on a single alter to broad, egalitarian allocations; and second, an individual’s social signature displays remarkable *intraindividual persistence* over time. Crucially, this persistence held even when personal networks underwent substantial turnover, prompting Saramäki et al. [10] to hypothesize a “slot-filling” dynamic in which individuals maintain stable cognitive slots (e.g., for high-bandwidth, medium-bandwidth, and low-bandwidth contacts) and replenish them with new alters as old relationships fitting a specific slot fade.

Over the past decade, a rich multidisciplinary literature has expanded upon Saramäki and colleagues’ initial formulation. Scholars have investigated social signatures across diverse communication channels, including text messaging [11], online collaborative platforms and social networking sites [12, 13], and in-person sensor networks [14, 15]. Physicists and network scientists have developed continuous resource allocation models [16] and generative micro-foundations rooted in the competition between cumulative advantage and random exploration [17], while methodological work has sought to disentangle genuine tie strength heterogeneity from network size stability [18, 19]. Recent inquiries have even leveraged signature distinctiveness as a behavioral biometric capable of re-identifying individuals in anonymized data [20].

Despite these theoretical, empirical, and modeling advances, fundamental open questions remain at the heart of the social signatures framework. First, there is the issue of timescale resolution as different researchers have used ad hoc criteria for observing social signatures. This leads to the question: Over what observational timescales do empirical signatures converge to stable, asymptotic parameters? Even more critically, because previous investigations have relied almost exclusively on commercial telecommunications metadata or passive digital traces, researchers have encountered

a severe *content black box*: commercial logs record the timing and frequency of calls or messages, but contain little information regarding dimensions of social support, emotional content, or the relational content of the underlying relationships. Does the steep drop-off between Rank 1 and lower-ranked alters correspond to content-based distinctions in social support, such as emotional comfort, instrumental advice, companionship, or financial aid? Furthermore, the behavioral mechanisms governing signature stability remain empirically unverified: does signature persistence truly reflect active “slot-filling,” or does it merely mirror the retention of core ties? Finally, we know little about what accounts for the observed deep interindividual variation in social signatures. What psychological dispositions and personality traits lie behind the strong patterns of interindividual variation observed producing steep versus flat signatures in the first place?

This study addresses these questions by analyzing data from *NetHealth*—a longitudinal cohort study tracking an incoming class of undergraduate students at the University of Notre Dame across two full academic years [21–24]. By pairing continuous, passive smartphone communication records that pool all outgoing channels logged by NetHealth—voice calls, SMS, MMS, and WhatsApp messages—($N = 515$ egos; 15,609,345 outgoing events, see Methods) with an academic calendar, eight waves of ego-network surveys (35,913 alter nominations), complete alter-alter structural edge lists (174,748 ties), and longitudinal psychometric batteries (Big Five personality traits), we build a comprehensive empirical investigation of the structure, correlates, and determinants of inter- and intraindividual variation in social signatures. Pooling channels is a deliberate design choice: voice calls account for only 2.3% of all communication events logged in NetHealth, versus 97.7% for text-based channels combined (87.2% SMS, 7.3% WhatsApp, 3.2% MMS), so restricting the analysis to calls alone would characterize a numerically marginal sliver of students’ mediated communication rather than their overall relational investment. More specifically, we systematically evaluate signature persistence across multiple calendar and rolling timescales, test parametric power-law versus exponential scaling, establish the observation duration required for parameter convergence, ground communication ranks in multidimensional social support dimensions, test the slot-filling dynamic under top-alter turnover, and estimate the psychological and structural drivers of signature morphology.

The remainder of this paper is organized as follows. Section 2 reviews the intellectual origins and recent advances in the social signatures literature, identifying five critical theoretical and empirical frontiers that our study addresses. Section 3 presents the empirical baseline results on persistence, power-law scaling, and parameter convergence. Section 4 develops four substantive expansions: social support grounding for alters in different slots, the slot-filling mechanism, multilevel panel models of signature divergence, and personality determinants of signature shape. Section 5 discusses the theoretical implications of our findings for evolutionary anthropology, sociology, and computational social science. The *NetHealth* dataset, temporal-window binning schemes, and mathematical formulations of social signatures and divergence metrics are described in Methods.

2 Prior Work and Open Questions

The concept of the social signature sits at the intersection of evolutionary anthropology, social network analysis, and computational network science. In this section, we review the evolution of this framework over the past decade, synthesizing findings from mobile telecommunications, online interaction platforms, generative physical models, and methodological refinements. We then articulate five foundational frontiers left open in the prior literature that the *NetHealth* study is uniquely positioned to address.

2.1 Cognitive Constraints and Persistence

The theoretical foundation of the social signatures framework is rooted in Dunbar’s influential social brain hypothesis [4, 5, 7], which posits that the evolution of large primate brains was driven by the cognitive demands of managing complex social relationships. Neocortical processing capacity imposes an upper constraint on the number of personal ties an individual can actively maintain, a ceiling widely recognized as Dunbar’s number (≈ 150). Within this outer boundary, egocentric personal networks are organized into a nested hierarchy of concentric layers or circles, where group sizes scale by a factor of approximately 3: an intimate support clique of 1 to 2 primary partners, a sympathy group of roughly 5 close ties, an affinity group of 12 to 15 core friends, and broader layers of 50 and 150 acquaintances [8, 9]. Crucially, maintaining social ties requires not only cognitive processing power but also the expenditure of finite temporal and emotional resources [6]. Because time is a non-renewable, zero-sum budget, individuals cannot allocate attention uniformly across their contacts; investing heavily in one relationship inevitably constrains the bandwidth available for others.

In a foundational paper, Saramäki et al. [10] formalized this resource allocation process by introducing the concept of *social signatures*. Rather than treating personal networks as unweighted graphs or simple categorical layers, Saramäki and colleagues conceptualized an ego’s relational environment as a continuous allocation profile. For an ego i maintaining contacts with K_i alters over an observation window w , alters are sorted in descending order of communication effort (measured by contact frequency or total call duration), $w_{i,(1),w} \geq w_{i,(2),w} \geq \dots \geq w_{i,(k_i),w}$. The social signature is defined as the normalized vector of interaction proportions directed to each rank $r \in \{1, \dots, k_i\}$:

$$p_{i,w}(r) = \frac{w_{i,(r),w}}{\sum_{j=1}^{k_i} w_{i,(j),w}}, \quad \text{such that} \quad \sum_{r=1}^{k_i} p_{i,w}(r) = 1. \quad (1)$$

To compare signatures across time or between different individuals, Saramäki et al. [10] used the Jensen-Shannon Divergence (JSD) with zero-padding on-the-fly [25]. For two signatures P and Q , zero-padded to common length $K = \max(|P|, |Q|)$, the distance is:

$$\text{JSD}(P \parallel Q) = H(M) - \frac{1}{2}[H(P) + H(Q)], \quad (2)$$

where $M = \frac{1}{2}(P + Q)$ is the midpoint distribution and $H(\cdot)$ denotes the Shannon entropy in bits, $H(P) = -\sum_{r=1}^K p_r \log_2(p_r)$.

Tracking 24 high school graduates transitioning into university over 18 months, Saramäki et al. [10] established two key empirical generalizations. First, individuals display marked *interindividual* diversity in their communication profiles: while all signatures exhibit heavy-tailed skewness, some egos maintain hyper-concentrated distributions directing over 40% of their calls to Rank 1, whereas others distribute contact effort much more evenly across top alters. Second, an individual’s signature displays persistent stability over time, such that self-divergence between an ego’s signatures across consecutive observation periods (d_{self}) is consistently smaller than the reference divergence between random pairs of individuals (d_{ref}).

Even more remarkably, signature persistence endures despite high turnover in network membership: between high school and university, participants replaced between 60% and 80% of their active contacts (Jaccard distance $1 - J \approx 0.60\text{--}0.80$). This striking observation led Saramäki et al. [10] to hypothesize the existence of a “slot-filling” mechanism: individuals do not adapt their overall allocation profile to the identities of their current alters; rather, they maintain pre-existing relational slots and dynamically slot newly acquired alters into vacated positions.

2.2 Multichannel Communication

Following Saramäki and colleagues’ initial findings, subsequent research examines whether social signatures are idiosyncratic to traditional voice communication via phone or reflect a universal organizing principle across contemporary digital communication channels.

First examining commercial mobile telecommunications at large scale, Heydari et al. [11] analyzed voice call and Short Message Service (SMS) records from a European carrier spanning more than half a million egos. They confirmed that the signature persistence observed by Saramäki et al. [10] in a small student cohort held across nationwide populations. Crucially, Heydari et al. [11] showed that text messaging signatures also exhibited intraindividual persistence over time ($d_{\text{self}} < d_{\text{ref}}$). Furthermore, when comparing call and SMS signatures for the same ego, they found a moderate structural resemblance, even though the specific alters contacted via voice calls and text messages showed minimal overlap.

Broadening the inquiry to online environments, Li et al. [12] investigated the dynamical evolution of social signatures among users of Facebook Wall posts and Wikipedia Talk pages across multiple consecutive intervals. Their empirical findings confirmed that social signatures remain stable in collaborative online networks despite continuous member churn. Extending the analytical framework, Li et al. [12] introduced the concept of “structural signatures” based on topological embeddedness (the number of mutual friends shared between ego and alter in the global network). They showed that alters ranked by structural embeddedness produce rank-frequency profiles that mirror interaction-based social signatures, showing that behavioral communication intensity reflects positions within the wider community structure.

In a comprehensive multi-modal study, Li and Bond [14] analyzed high-resolution data from the

Copenhagen Networks Study [26], tracking more than 700 university students across three distinct interaction modes: mobile phone calls, text messages, and Bluetooth-sensed in-person physical proximity. Li and Bond [14] showed that social signatures were persistent across time and consistent across all three communication channels. However, they uncovered key modality-specific differences: communication networks mediated by voice calls and text messages were substantially more stable than in-person proximity networks, which exhibited higher turnover and greater susceptibility to circumstantial fluctuations. Drawing on media richness theory [27], Li and Bond [14] argued that individuals match the richness and synchronicity of specific communication channels to the ambiguity of relational tasks, choosing lean mediated channels for core ties while using face-to-face contact for broader social exploration.

More recently, Liu et al. [28] evaluated multi-channel mobile communications using entropy evaluation methods and null models. Their analyses revealed an empirical U-shaped trade-off between calling and texting as an individual’s total contact portfolio expands, alongside clear vitality ceilings: egos maintained active communication vitality up to approximately 10 contacts in voice calling, 100 in text messaging, and 20 in hybrid communication. Together, these studies confirm that social signatures represent an enduring, cross-modal behavioral pattern, but also indicate that different communication media capture distinct slices of an individual’s total relational life.

2.3 Theoretical Models and Scaling

In parallel with empirical extensions across platforms, theorists and statistical physicists sought to identify the mathematical laws and generative mechanisms governing social signatures. A prominent debate concerned whether personal networks are best understood as discrete, stepwise layers (the classical Dunbar circles) or as smooth, continuous distributions.

Bridging this divide, Tamarit et al. [16] formulated a statistical-mechanical model of resource allocation based on the principle of maximum entropy. Rather than categorizing ties into a small number of arbitrary discrete bins, in which the cost of interaction is assumed to be uniform within each bin, Tamarit and colleagues treated relational maintenance costs as continuous variables. Their continuum model showed that discrete Dunbar circles and continuous social signatures are mathematically unified manifestations of a single resource allocation process under finite cognitive capacity. Under this formulation, they predicted and empirically identified a universal scaling parameter, $\eta \approx 6$, representing the ratio of relationship costs between adjacent levels. Evaluating datasets spanning mobile phone logs, face-to-face contacts, and Facebook interactions, Tamarit et al. [16] showed that the distribution of estimated parameters across individuals clustered tightly around this universal value.

2.4 Disentangling Degree and Tie Strength

As empirical work on social signatures expanded, methodologists noted an important vulnerability in how social signature persistence had historically been measured. In a rigorous examination, Heydari et al. [18] and Heydari [19] pointed out that an ego’s social signature contains two distinct

structural features: its length (the personal network size or degree, k) and the internal distribution of interaction weights across those ranks (tie strength heterogeneity).

Standard evaluations of signature persistence calculate the Jensen-Shannon Divergence or Euclidean distance between zero-padded signature vectors. However, Heydari et al. [18] showed that JSD distance is mathematically sensitive to differences in signature length: two signatures of similar length will mechanically yield a lower divergence than two signatures of disparate lengths, regardless of their internal curvature. Because an individual’s network degree tends to be relatively stable across consecutive observation windows, the self-divergence (d_{self}) is systematically calculated between signatures of similar length. Conversely, the reference divergence (d_{ref}) compares random pairs of individuals in the population, who often exhibit substantial differences in degree. Consequently, standard JSD comparisons risk conflating the persistence of network size with the persistence of communication allocation strategy.

To resolve this confounding, Heydari et al. [18] evaluated degree-insensitive metrics of tie strength heterogeneity, including the Gini coefficient distance:

$$\Delta G(G_1, G_2) = |G_1 - G_2|, \quad \text{where} \quad G = \frac{2}{k} \frac{\sum_{r=1}^k r \cdot w_r}{\sum_{r=1}^k w_r} - \frac{k+1}{k}, \quad (3)$$

and the normalized alter-preferentiality distance derived from Iñiguez and colleagues’ generative model:

$$\Delta\beta(\beta_1, \beta_2) = \frac{|\beta_1 - \beta_2|}{\beta_1 + \beta_2}. \quad (4)$$

Applying these degree-invariant measures to large-scale mobile phone datasets, Heydari et al. [18] proved that the persistence of personal network structure cannot be attributed to degree stability alone. Even after strictly controlling for degree differences, intraindividual distances remained significantly smaller than interindividual reference distances. This confirmed that tie-strength heterogeneity is an authentic, degree-independent behavioral trait of egos: individuals possess an enduring personal allocation style (whether concentrated or egalitarian) that persists regardless of the number of contacts they maintain.

2.5 Longitudinal Stability and Privacy

Finally, two recent developments have examined the longitudinal boundaries of signature persistence and its implications for individual privacy.

Investigating the temporal stability of social signatures over extended observation horizons, Koltsova et al. [13] tracked private messaging on the Russian social networking site *Vkontakte* (VK) across 18 consecutive months. While replicating the baseline finding that short-term self-distances were smaller than interindividual reference distances ($d_{\text{self}} < d_{\text{ref}}$), Koltsova et al. [13] made a critical new discovery: intraindividual signature changes across longer spans (e.g., between Months 1–6 and Months 13–18) were statistically significant via Kolmogorov-Smirnov tests. They argued that social signatures are not immutable person-level constants; rather, they undergo subtle developmental drift

as individuals experience shifting life circumstances, suggesting that signature stability is bounded by long-term life transitions.

Conversely, the very uniqueness and temporal persistence of personal allocation profiles has raised significant data privacy concerns. Showing this potential for behavioral re-identification, Jia et al. [20] examined how multidimensional social signatures constructed from interaction timing, contact volume, and communication ranks can de-anonymize users in supposedly anonymous datasets. Using low-sensitivity interaction metadata from email, call logs, and app engagement records, Jia et al. [20] achieved user re-identification accuracies of up to 87%. Their findings establish that an individual’s communication signature is so distinct and persistent that it operates as a behavioral biometric fingerprint, challenging conventional assumptions regarding the safety of anonymized network data under privacy regulations such as the GDPR and CCPA.

2.6 Open Questions and Frontiers

Despite the substantial progress achieved across these studies of the social signatures phenomenon, the literature has reached an empirical impasse defined by five major open questions that existing datasets cannot answer.

First, prior literature encounters a *persistent content black box*. Commercial telecommunications records [10, 11, 17], online messaging logs [12, 13], and Bluetooth sensor traces [14] are completely devoid of relational content [29]. Consequently, researchers have been unable to determine what communication ranks actually mean in network theory terms: does the steep drop in communication effort between Rank 1 and lower-ranked alters reflect mere quantitative differences in contact volume, or does it mark transitions between distinct dimensions of social support [30] such as emotional confiding, instrumental advice, social companionship, financial aid, or kinship versus peer friendship? While Koltsova et al. [13] examined a subjective closeness rating on a small sample of 39 users, no study has linked large-scale communication ranks to multidimensional sociometric support taxonomies. The *NetHealth* study can help address this impasse by cross-referencing $N = 21,273$ communication-ranked dyads, pooled across voice calls, SMS, MMS, and WhatsApp messages, with eight waves of longitudinal ego-network surveys that record multidimensional social support and tie categories, showing that communication rank tracks sharp, content-based divisions in social support across classic Dunbar tiers.

Second, the behavioral mechanism underlying signature stability requires direct empirical verification. While Saramäki et al. [10] and Heydari [19] hypothesized that signature stability persists because individuals slot new alters into pre-existing relational positions, they lacked the alter-level tracking and longitudinal survey depth required to verify this mechanism. Specifically, does an individual’s signature remain stable because their core ties (Rank 1 and Rank 2) never change, or does stability truly endure when an individual experiences complete turnover of their primary confidant? Leveraging continuous passive logs of pooled multichannel communication across eight academic semesters with an average semester alter turnover of 73.5%, we explicitly separate transitions where an ego retains their primary alter from transitions where the primary alter is

completely replaced. We show that even under complete turnover of the top confidant—which occurs in the majority (59.5%) of semester transitions once texting is pooled with calls—signature divergence shifts only modestly (from 0.041 to 0.083), providing direct empirical confirmation of the slot-filling hypothesis.

Third, the psychological foundations of signature curvature remain unmapped. Theoretical models of personal network evolution [7, 15, 17] assume that individual differences in alter-preferentiality (β) and signature decay exponents (α) reflect internal cognitive and affective dispositions. Yet commercial telecommunications providers have no psychometric data on their subscribers, and university sensor studies [14] did not model personality as a predictor of signature steepness. Beyond preliminary exploratory correlations [31], no study has rigorously estimated the extent to which validated personality dimensions govern the curvature of social signatures. We link objective communication decay exponents (α_i) to longitudinal psychometric batteries measuring the Big Five personality traits. Through regression modeling, we show that Negative Emotionality predicts steeper, more concentrated signatures ($\beta = 0.076, p = 0.0016$), revealing that heightened negative emotionality is associated with somewhat greater reliance on a narrow core of alters, albeit a more modest association than observed when voice calls are analyzed in isolation.

Fourth, the field lacks empirical guidance on multi-timescale hierarchies and parameter burn-in convergence. Prior empirical investigations evaluated social signatures at arbitrary, isolated temporal resolutions: Li and Bond [14] examined 1-month intervals, Heydari et al. [11] used 6-month windows, and Saramäki et al. [10] examined 9-month terms. However, no study has evaluated how signature stability behaves across a nested hierarchy of timescales within the same population. More crucially, prior literature offers no guidance on parameter burn-in: how long must an individual be continuously observed before their estimated signature decay parameter α converges to its true asymptotic value? We evaluate social signatures across eight temporal window schemes spanning calendar definitions (academic years, semesters, quarters, months) and rolling weekly intervals (3-week rolling, 2-week discrete, 1-week discrete). Furthermore, by tracking cumulative observation intervals from 2 to 24 months, we establish an empirical burn-in threshold showing that personal signature parameters stabilize within 6 to 8 months of continuous observation.

Finally, personal communication signatures must be situated within their broader network structural embedding. Previous studies have modeled communication signatures in isolation from the structural connectivity of the surrounding network. Yet communication is embedded within broader community structures, residence halls, and friendship groups [12]. How do alter turnover, ego network transitivity (clustering), and shifts in overall communication volume jointly govern signature stability over time? *NetHealth* pairs continuous individual phone logs with 174,748 alter-alter structural ties connecting peers within each respondent’s personal egocentric network across survey waves. We estimate multilevel linear mixed-effects models predicting signature self-divergence (d_{self}) across consecutive semesters as a joint function of dyadic Jaccard turnover, triadic clustering, and communication volume shifts, providing the first structural panel evaluation of signature self-divergence.

3 Results: Persistence and Scaling

3.1 Signature Morphology

Figure 1 displays empirical social signatures averaged across egos for ranks 1 through 15 across temporal window definitions. Across all timescales, communication effort exhibits pronounced skewness. In semester windows, the single top-ranked alter ($r = 1$) commands an average of 35% of an ego’s total pooled communication effort, while the second- and third-ranked alters capture an additional 12% and 8%, respectively (top three alters: 55% of total volume). Beyond rank 5, communication drops to a thin, prolonged tail of occasional contacts.

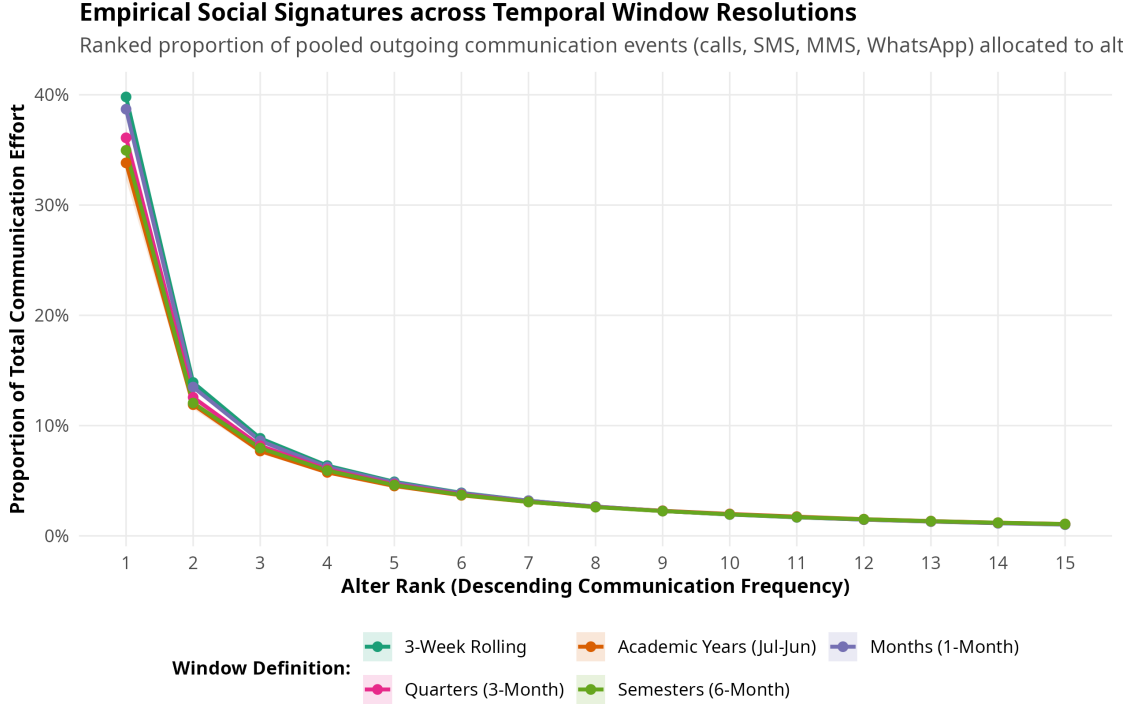


Figure 1: Empirical Social Signatures across Temporal Window Resolutions (Ranks 1–15). Ribbons denote 95% confidence intervals around the mean interaction proportion.

3.2 Persistent Individual Differences Across Timescales

Supplementary Table S2 reports the formal statistical comparison between intraindividual self-divergence (d_{self}) and interindividual reference divergence (d_{ref}) across all eight window schemes.

Across every temporal resolution examined, intraindividual self-divergence is significantly lower than interindividual reference divergence ($p < 0.0001$, Wilcoxon signed-rank test). In semester windows, mean self-divergence is 0.0658, compared to a reference divergence of 0.1324 ($V = 3602$, $p < 0.0001$). The persistence finding is, if anything, sharper at fine-grained weekly scales once text messages are pooled in: in 3-week rolling windows, an individual’s signature remains exceptionally consistent across time ($d_{\text{self}} = 0.0182$) relative to the broader population ($d_{\text{ref}} = 0.1379$, $p < 0.0001$),

reflecting the much larger daily event volume available to stabilize short-window estimates once texting is included. As shown in Figure 2, the distributions of self and reference divergence exhibit minimal overlap, confirming that pooled personal communication signatures reflect enduring individual traits, not an artifact specific to voice-calling behavior.

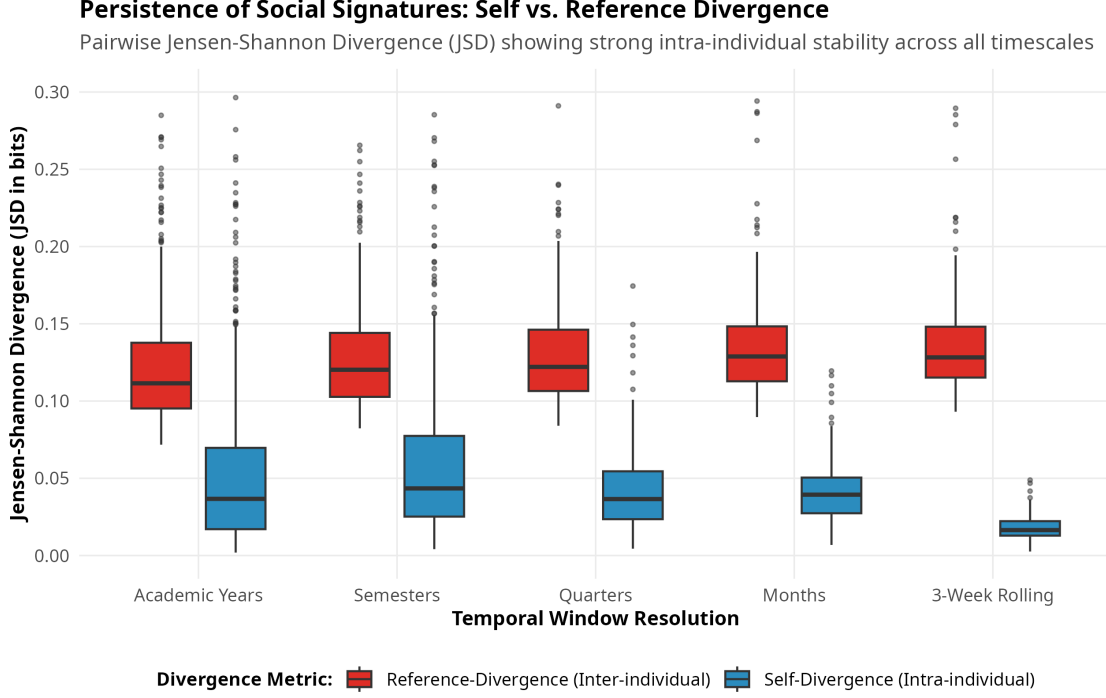


Figure 2: Persistence of Social Signatures: Self-Divergence vs. Reference-Divergence across temporal resolutions. Boxplots show the distribution of individual ego means.

3.3 Parametric Model Evaluation

We evaluated two competing functional specifications for the signature curve:

$$\text{Power-Law: } p(r) = c \cdot r^{-\alpha} \iff \log p(r) = \log c - \alpha \log r \quad (5)$$

$$\text{Exponential: } p(r) = c \cdot e^{-\beta r} \iff \log p(r) = \log c - \beta r \quad (6)$$

As summarized in Supplementary Table S3 and Figure 3, the power-law model continues to fit pooled signatures well, but its advantage over the exponential specification narrows once text messages are pooled in with calls. In semester windows, the power-law model achieves a mean R^2 of 0.933, essentially matching the exponential specification’s mean R^2 of 0.900, and is preferred by Akaike Information Criterion (AIC) in only 60.8% of models (versus 97.0% under calls alone); in monthly windows the two specifications are statistically indistinguishable on average ($R^2 = 0.912$ vs. 0.911; power-law preferred in 47.8% of models). The power-law decay exponent α averages 1.94 (SD = 0.33) in semester windows, noticeably steeper than the calls-only estimate ($\alpha \approx 1.2$),

consistent with the pooled signature’s more concentrated top rank once high-volume but low-salience texting exchanges are folded into the same metric as calls.

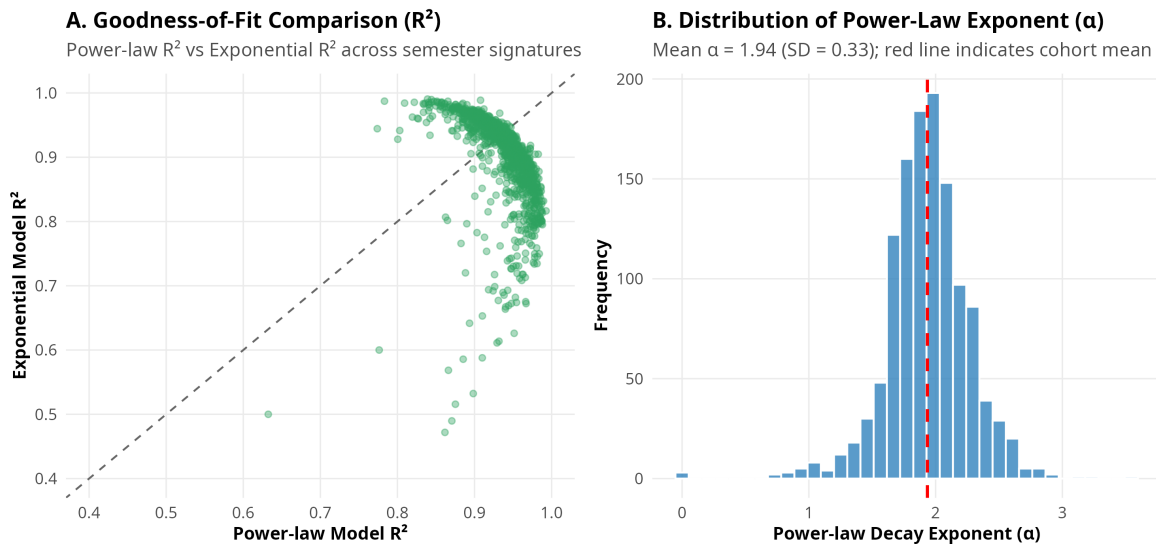


Figure 3: Power-Law vs. Exponential Model Fits. (A) Scatterplot of R^2 values comparing power-law against exponential fits across semester signatures. (B) Empirical distribution of power-law decay exponent (α).

3.4 Parameter Burn-In Convergence

To determine how long an individual must be observed before their estimated signature parameters stabilize, we tracked the common cohort across cumulative observation periods ranging from 2 to 24 months (Supplementary Figure S1). At two months of observation, the mean estimated exponent is $\alpha = 1.90$, with a mean absolute error of 0.312 relative to the 24-month asymptotic value. By four months, the mean absolute error declines to 0.230 ($R^2 = 0.925$). The mean absolute error drops below 0.10 around month 9 (0.099, $R^2 = 0.940$) and below 0.05 by month 16 (0.051, $R^2 = 0.949$). These findings show that roughly nine to ten months of continuous digital observation is needed for parameter convergence once the pooled, higher-volume multichannel signal is used, a somewhat longer burn-in than the six-to-eight-month threshold observed for voice calls alone, likely because the steeper, more concentrated pooled signatures take longer to separate from early-window sampling noise in the long tail of infrequently texted alters.

4 The Relational Content of Social Signatures

4.1 Role Relations and Dimensions of Social Support

While prior work on social signatures has treated communication ranks as purely behavioral tallies, we linked the pooled communication records of the collegiate semester cohort ($N = 307$ egos) to the longitudinal network surveys. This linkage matched $N = 21,273$ communication-ranked dyadic

observations across 300 unique egos to relationship categories and dimensions of social support across six theoretical Dunbar tiers (Table 1 and Supplementary Figure S2).

Table 1: Relational Composition and Support Functions Across Signature Rank Tiers ($N = 21,273$ Dyads)

Rank Tier	Ties (N)	% Family	% Friend	% Emotional	% Advice	% Companionship	% Financial
Rank 1 (Core)	1,003	23.3%	53.1%	80.8%	82.3%	86.3%	26.1%
Ranks 2–3	1,877	36.1%	61.9%	74.4%	77.9%	82.6%	30.0%
Ranks 4–5	1,762	33.5%	65.2%	68.8%	74.3%	81.5%	24.0%
Ranks 6–10	3,703	28.0%	71.0%	58.4%	63.7%	77.7%	17.8%
Ranks 11–20	4,934	18.5%	79.3%	44.9%	49.9%	71.7%	9.2%
Ranks > 20	7,994	11.3%	83.9%	28.1%	32.5%	59.3%	4.0%

Pooling all communication channels substantially revises the relational portrait of the primary signature slot. Rather than the kinship-dominated hub observed under voice calls alone, Rank 1 under pooled communication is majority friend (53.1%) and only secondarily family (23.3%); family representation is not monotonic across ranks but instead peaks at Ranks 2–3 (36.1%) before declining steadily toward the periphery (11.3% beyond Rank 20), consistent with parents and other kin occupying a secondary rather than primary position once high-frequency peer texting is counted alongside calls. Friend representation rises monotonically from 53.1% at Rank 1 to 83.9% in the outermost tier. Despite this compositional reversal, the support gradient retains the same qualitative shape documented under calls alone: emotional support (80.8% at Rank 1) and advice (82.3% at Rank 1) decline monotonically across tiers (to 28.1% and 32.5% for outer ties), while companionship remains comparatively elevated throughout (86.3% at Rank 1, 81.5% at Ranks 4–5, and still 59.3% beyond Rank 20), indicating that peripheral communication continues to serve social and recreational integration functions even as its relational composition shifts toward friends. Financial support is substantially lower across all tiers than under calls alone (26.1% at Rank 1 versus 64.1% previously), consistent with financial assistance being disproportionately a voice-call-mediated (often parent-initiated) function that is diluted once pooled with high-volume peer texting.

4.2 Evaluative and Cognitive Alignment: Closeness, Duration, and Salience

Beyond discrete relational content divisions based on role relations or social support, an alter’s position slot in the social signature is strongly and systematically correlated with primary evaluative and cognitive dimensions of the social tie (Figure 4 and Supplementary Table S4).

First, subjective closeness [32] exhibits a steep gradient across signature tiers ($F(5, 21267) = 499.2, p < 0.0001$): 93.6% of alters occupying the Rank 1 slot are rated as “especially close,” declining steadily to 81.6% at Ranks 4–5, and 45.3% at Ranks > 20 ($r = 0.201, \rho = 0.336$ with communication proportion, $p < 0.0001$; Figure 4A). Second, tie duration, or relationship age [33], shows a non-monotonic pattern across tiers ($F(5, 21267) = 194.5, p < 0.0001$) that mirrors the hump-shaped family-composition gradient documented above: Rank 1 alters average 6.6 years known (median 2.5 years), rising to a peak of 8.9 years at Ranks 2–3 (median 5 years), before declining to 5.0

years (median 2.3 years) beyond Rank 20; because this relationship is non-monotonic, its simple bivariate correlation with communication proportion is weak ($r = 0.027, \rho = 0.164, p < 0.0001$), even though the tier-level differences are highly significant (Figure 4B). Third, social signature position corresponds closely to cognitive salience, operationalized as the alter’s recall order during survey elicitation ($F(5, 21267) = 882.1, p < 0.0001$): 88.3% of Rank 1 alters are recalled within the top 5 positions (mean recall position 2.5), whereas alters in outer tiers average a recall position of 9.1 with only 31.3% recalled in the top 5 ($r = 0.262, \rho = 0.436$ with communication proportion, $p < 0.0001$; Figure 4C).

Evaluative and Cognitive Alignment Across Social Signature Rank Tiers

Closeness, relationship duration, and recall salience across pooled communication signature rank tiers

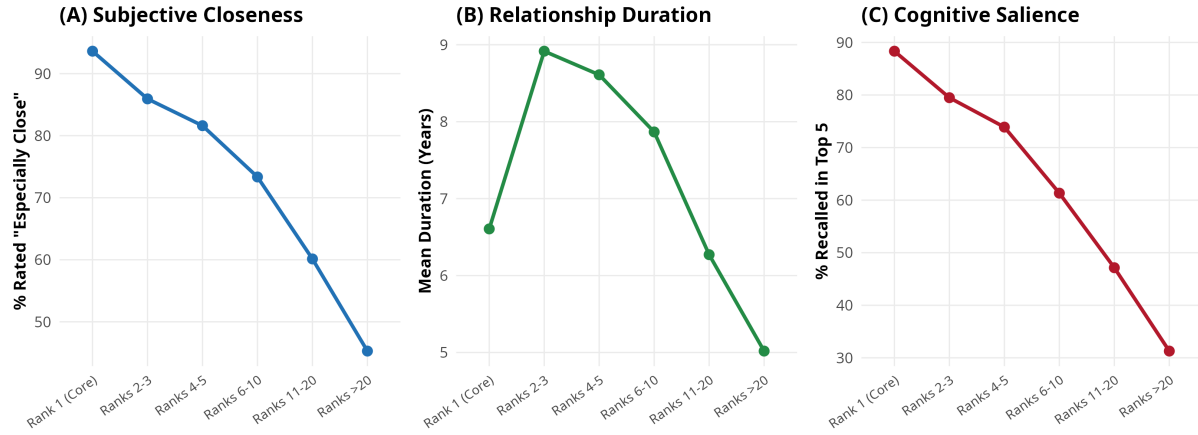


Figure 4: Evaluative and Cognitive Alignment Across Social Signature Rank Tiers ($N = 21,273$ Dyad-Window Observations). (A) Subjective Closeness: percentage of alters rated as “especially close.” (B) Relationship Duration: mean relationship longevity in years. (C) Cognitive Salience: percentage of alters recalled within the top 5 positions during survey elicitation.

Because personal ties are hierarchically nested within egos, and a given dyad can appear in more than one semester window, pooled regression risks inflating test statistics due to correlated errors across observations belonging to the same participant. Supplementary Table S4 therefore reports linear mixed-effects models with ego random intercepts, estimated on the complete-case dyad-window sample ($N = 16,496$ observations across $J = 295$ egos) to properly partition between-ego and within-ego variance. In the baseline evaluative model (Model 1), the ego-level intraclass correlation (ICC) is 0.065, indicating that roughly 7% of the variance in communication allocation is attributable to unobserved ego-level baselines (e.g., overall communication volume and network size). Substantively, subjective closeness ($t = 17.9, p < 0.0001$) and interpersonal trust ($t = 8.3, p < 0.0001$) both strongly and positively predict communication proportion. Adding tie duration in Model 2 reveals a negative independent effect once closeness and trust are held constant ($t = -11.1, p < 0.0001$): among ties of comparable closeness and trust, longer-known alters receive a smaller share of pooled communication, consistent with durable kin ties being maintained through sparser, higher-investment contact while shorter-duration college friendships are sustained through high-frequency texting. Model 3 incorporates cognitive salience, showing that top-of-mind recall accessibility is the strongest

single predictor of allocation ($t = 21.5, p < 0.0001$), while closeness ($t = 6.7$), trust ($t = 9.3$), and duration ($t = -11.3$) each retain significant effects. Finally, Model 4 shows that these attributes jointly govern an alter’s ordinal signature position ($p < 0.0001$ for closeness, duration, and salience; $p = 0.014$ for trust), with greater closeness, recall salience, and (net of closeness and trust) shorter duration each pulling alters closer to the primary core slot (Rank 1). Thus, under pooled multichannel communication, signature rank remains tightly coupled to cognitive salience and subjective closeness, but its relationship to relationship duration reverses sign once high-frequency, low-duration peer texting is counted alongside sparser, long-duration kin contact.

4.3 The Slot-Filling Mechanism Under Alter Turnover

A central theoretical puzzle in social signatures research is how an ego’s signature can remain stable despite the reality of frequent alter replacement. To investigate how frequent alter replacement is in the *NetHealth* data, we calculated the dyadic Jaccard turnover between consecutive semesters:

$$\text{Turnover}_{i,w} = 1 - \frac{|A_{i,w} \cap A_{i,w+1}|}{|A_{i,w} \cup A_{i,w+1}|} \quad (7)$$

where $A_{i,w}$ denotes the set of active alters for ego i in window w .

Under pooled communication, *alter turnover between consecutive semesters averages 73.5%* ($SD = 0.088$) across $N = 921$ ego-transition pairs. Despite replacing roughly three out of four alters every six months, egos maintain an average self-divergence of only 0.0658 ($r = 0.386, p < 0.0001$ between turnover and self-divergence). The single top-ranked alter is retained in only 40.5% of consecutive semester intervals—a considerably lower retention rate than under voice calls alone (59.1%), consistent with the identity of a student’s single highest-volume contact (often a new roommate, partner, or close friend reached primarily by text) changing more readily than the identity of their most frequent caller. Even so, when the top alter is replaced, the overall self-divergence increases only from 0.041 to 0.083 (Figure 5)—roughly a doubling, similar in relative magnitude to the shift observed under calls alone, even though complete top-alter replacement is now the more common case.

This pattern provides direct empirical confirmation of the *slot-filling hypothesis*, and if anything strengthens it: even when the identity of the single most-contacted alter changes in the majority of semester transitions, an individual’s overall allocation curve shifts only modestly, indicating a predefined cognitive template for relationship allocation into which new social entrants are seamlessly slotted.

4.4 Multilevel Panel Models of Signature Divergence

To identify the structural and psychological drivers of signature mutation over time, we estimated linear mixed-effects panel models with ego random intercepts using the `lme4` package [34] in R [35]:

$$\text{JSD}_{it} = \beta_0 + \beta_1 \text{Turnover}_{it} + \beta_2 \Delta \text{Activity}_{it} + \mathbf{X}_{it} \boldsymbol{\gamma} + u_i + \epsilon_{it} \quad (8)$$

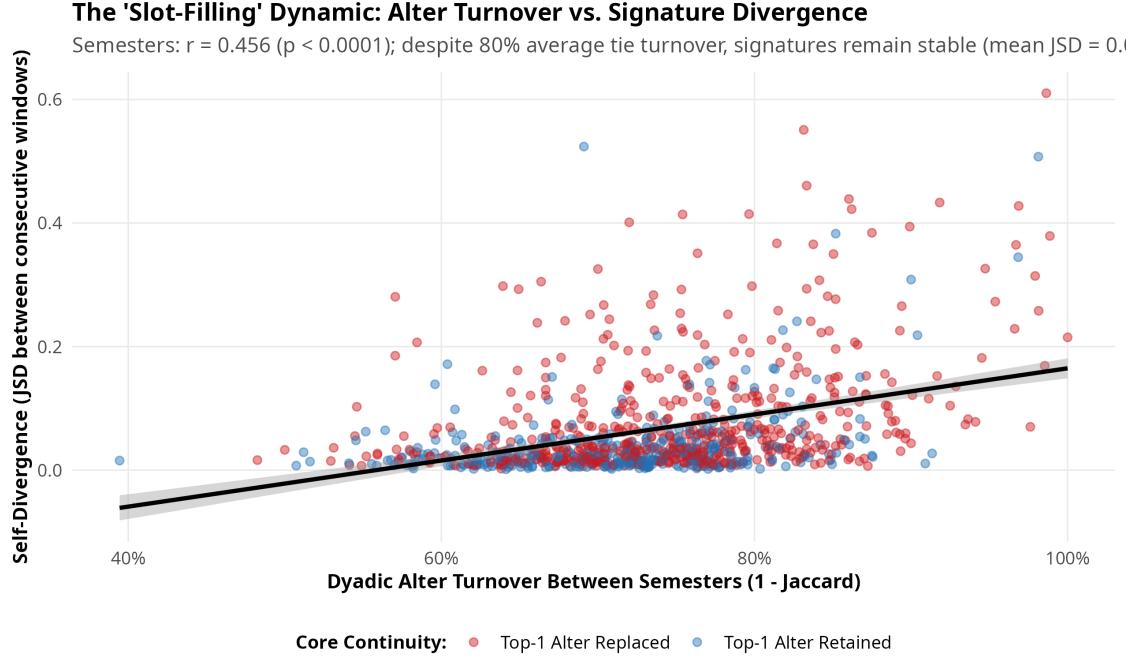


Figure 5: The ‘Slot-Filling’ Dynamic: Alter Turnover vs. Signature Divergence ($r = 0.386, p < 0.0001$). Blue points denote intervals where the Rank 1 alter was retained; red points denote intervals where the Rank 1 alter was replaced.

where $u_i \sim \mathcal{N}(0, \sigma_u^2)$ and $\epsilon_{it} \sim \mathcal{N}(0, \sigma_\epsilon^2)$.

As reported in Supplementary Table S5, alter turnover remains the primary driver of signature divergence ($\beta = 0.349, p < 0.0001$ in the integrated model), a coefficient essentially unchanged from the calls-only estimate ($\beta = 0.351$). Large shifts in pooled communication volume also significantly predict divergence, though the coefficient is smaller in absolute terms given the much larger scale of pooled event counts relative to calls alone. Net of turnover, ego network clustering shows only a marginal, non-robust association with signature stability ($p = 0.075$ in the integrated model), underscoring that signatures continue to reflect person-level relational habits rather than the triadic closure of the surrounding peer group. Baseline Negative Emotionality continues to predict lower self-divergence (greater stability) net of turnover and activity change ($\beta = -0.018, p = 0.012$), while Extraversion shows no significant association.

4.5 Personality Determinants of Signature Shape

What psychological dispositions generate steep (core-concentrated) versus flat (diffuse) social signatures? Regressing ego-mean power-law decay exponents (α_i) on Big Five personality traits and personal network degree (Supplementary Table S6 and Figure 6) shows that Negative Emotionality remains a statistically significant positive predictor of signature steepness once all communication channels are pooled ($\beta = 0.076, t = 3.19, p = 0.0016$), though both the coefficient and the overall model fit are noticeably attenuated relative to the calls-only analysis ($\beta = 0.098, p < 0.0001$;

$R^2 = 0.124$ versus 0.058 here). As shown in Figure 6A, individuals higher in baseline Negative Emotionality still exhibit somewhat steeper power-law decay exponents on average.

Figure 6B plots empirical pooled signatures across alter ranks 1 through 10, contrasting students at the high versus low extremes of Negative Emotionality (T3 vs. T1). Unlike the calls-only analysis, the two group-mean curves now overlap closely at the top ranks (Rank 1: 34.3% for the high group versus 34.9% for the low group; Rank 2: 12.6% versus 11.2%), indicating that the aggregate association between Negative Emotionality and α_i is diffuse across the full decay curve—driven by differences further into the tail of the distribution—rather than visible as a sharp gap in how much communication effort is directed to the single primary alter. Extraversion again shows no statistically significant association with decay steepness ($\beta = -0.026, p = 0.224$). Notably, Agreeableness no longer shows the modest positive association with concentration observed under calls alone ($\beta = -0.014, p = 0.659$, compared with $\beta = 0.063, p = 0.014$ previously), and conscientiousness and openness again show no detectable effects.

Negative Emotionality as a Driver of Egocentric Relational Concentration

Higher Negative Emotionality predicts significantly steeper power-law decay (bivariate $\beta = 0.084, p = 0.0002$); group-mean rank curves (panel B) overlap closely near the top ranks, indicating the association is diffuse across the full decay curve rather than concentrated at Rank 1-2

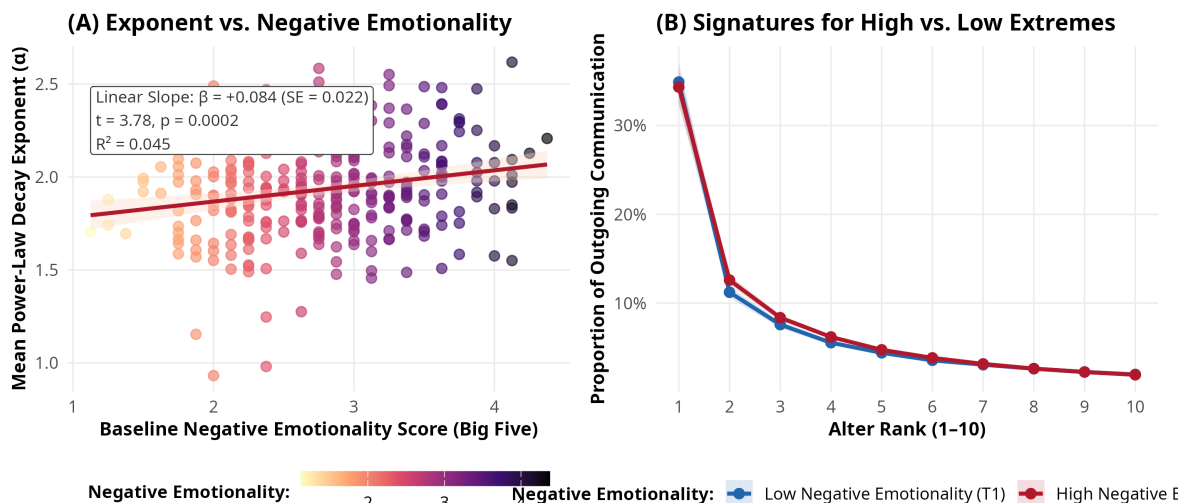


Figure 6: Negative Emotionality as a Driver of Egocentric Relational Concentration. (A) Scatterplot and bivariate linear regression of mean power-law decay exponent (α_i) on baseline Negative Emotionality score. (B) Mean empirical pooled signatures across alter ranks 1 through 10 contrasting high versus low Negative Emotionality extremes (T3 vs. T1); the two curves overlap closely at the top ranks.

While linear regression provides an omnibus test of parametric associations, it assumes additive relationships that may overlook non-linear thresholds or complex psychological interactions. To explore the hierarchical structure governing attention concentration, we trained a regression tree (Classification and Regression Tree; CART; Breiman et al. 36) predicting mean power-law decay exponents (α_i) from baseline Big Five traits (Supplementary Figure S3).

The pooled-data CART tree still roots on Negative Emotionality, splitting the cohort at a cutoff of 2.19 ($n = 306$). Participants below this threshold ($n = 56$) form a lower branch with a

reduced mean decay exponent ($\bar{\alpha} = 1.84$), while those above it ($n = 250$) subsequently split on Extraversion, Agreeableness, and Conscientiousness across several terminal leaves ranging from $\bar{\alpha} = 1.61$ to $\bar{\alpha} = 2.12$. However, the pooled-data tree’s overall predictive performance is weak: 10-fold cross-validated relative error exceeds 1.0 at every candidate split (minimum cross-validated error ≈ 1.01 at a single split, versus a training relative error of 1.00 at the root), indicating that none of the candidate splits reliably improve out-of-sample prediction beyond the sample mean. Relatedly, variable importance rankings—which, unlike the primary splits, also credit variables used only as surrogate splits—assign the largest share to Agreeableness (33%) rather than Negative Emotionality (20%), with Extraversion (24%), Conscientiousness (17%), and Openness (7%) also contributing. We therefore treat the non-linear, conditional moderation pattern identified in the calls-only CART analysis as not robust to the inclusion of text-based channels, and report the pooled-data tree here as a descriptive, exploratory structure rather than a validated predictive model.

5 Discussion

By linking continuous passive smartphone logs with multi-wave ego-network surveys and psychological batteries in the NetHealth cohort, this study provides empirical backing for social signature theory while advancing the paradigm into social network analysis and computational psychology. The findings resolve long-standing theoretical puzzles regarding the interplay between relational fluidity and cognitive constraint across life transitions.

In summary, we have shown that social signatures are an individual-level regularity that holds across multiple observational timescales, and that this regularity is not an artifact of restricting attention to voice calls: it replicates, and in several respects sharpens, once all outgoing communication channels (calls, SMS, MMS, and WhatsApp) are pooled into a single measure of communication effort. Across all eight temporal window definitions, intraindividual self-divergence (d_{self}) is systematically and significantly smaller than interindividual reference divergence (d_{ref} , $p < 0.0001$ across all resolutions). In collegiate semester windows, mean self-divergence is 0.0658, roughly half the reference divergence observed between random peers ($d_{\text{ref}} = 0.1324$). This stability persists not only at coarse-grained annual scales ($d_{\text{self}} = 0.0584$ versus $d_{\text{ref}} = 0.1255$) but also across fine-grained weekly intervals ($d_{\text{self}} = 0.0182$ versus $d_{\text{ref}} = 0.1379$ in three-week rolling windows). Formal nonparametric location tests confirm that pooled personal communication profiles reflect enduring behavioral traits rather than stochastic fluctuations.

Second, our parametric evaluations clarify the functional form of pooled social signatures, with an important qualification relative to the calls-only case. Power-law decay models ($p(r) = c \cdot r^{-\alpha}$, mean $R^2 = 0.933$) continue to fit the data well, but no longer decisively outperform exponential decay specifications ($R^2 = 0.900$); the power-law model is preferred by Akaike’s Information Criterion in only 60.8% of semester specifications, down from 97.0% under calls alone. The estimated power-law exponent averages $\alpha = 1.94$ (SD = 0.33), a steeper allocation profile than under calls alone ($\alpha \approx 1.2$), where the single top-ranked alter commands an average of 35% of total

communication effort. Longitudinal parameter burn-in tracking across 24 consecutive months shows that an ego’s estimated decay exponent converges to its asymptotic value within roughly nine to ten months of continuous digital observation (mean absolute error < 0.10), somewhat longer than the six-to-eight-month threshold observed for voice calls alone.

Third, our analysis helps open the content black box in previous work by linking 21,273 communication-ranked dyads to longitudinal ego-network survey nominations across six social signature Dunbar tiers. Pooling channels substantially revises the compositional portrait of the primary slot relative to calls alone: rather than a kinship-dominated hub, Rank 1 is majority friend (53.1%) and only secondarily family (23.3%), with family representation peaking instead at Ranks 2–3 (36.1%) before declining toward the periphery (11.3% beyond Rank 20). Despite this compositional reversal, the underlying support gradient retains its qualitative shape: emotional support (80.8% at Rank 1) and advice (82.3% at Rank 1) decline monotonically across tiers, while companionship remains comparatively elevated throughout (86.3% at Rank 1, still 59.3% beyond Rank 20), indicating that peripheral communication continues to serve social and recreational integration functions even as its relational composition shifts toward friends.

Fourth, linking communication ranks to survey evaluative measures shows that social signature position slots strongly correspond to cognitive accessibility and emotional closeness, with a notable reversal for relationship duration. In linear mixed-effects models accounting for the nesting of ties within egos and within semester windows ($N = 16,496$ dyad-window observations across 295 individuals, ICC = 0.065–0.079), cognitive recall salience ($t = 21.5, p < 0.0001$), subjective closeness ($t = 6.7\text{--}17.9, p < 0.0001$), and interpersonal trust ($t = 8.3\text{--}10.5, p < 0.0001$) each exert strong, independent positive effects on communication allocation. Tie duration, however, exerts a significant *negative* independent effect once closeness and trust are controlled ($t = -11.1$ to $-11.3, p < 0.0001$): among ties of comparable closeness and trust, longer-known alters receive a smaller share of pooled communication, consistent with durable kin ties being maintained through sparser, higher-investment contact while shorter-duration college friendships are sustained through high-frequency texting. Nearly 9 in 10 (88.3%) of alters in the primary slot are recalled within the first five positions elicited in survey name generators, confirming that behavioral communication ranks continue to reflect top-of-mind cognitive retrieval priority even as the underlying channel composition shifts.

Fifth, we provide direct empirical verification of the slot-filling mechanism, which if anything strengthens once texting is pooled with calls. Average semester-to-semester alter turnover is 73.5%, and egos preserve signature stability ($d_{\text{self}} = 0.0658$). The single top-ranked alter is retained in only 40.5% of consecutive intervals—lower than the 59.1% retention rate observed for voice calls alone—yet even when the primary contact is completely replaced by a new entrant, self-divergence increases only from 0.041 to 0.083. That signature stability persists even as the identity of the single most-contacted alter changes in the majority of transitions is a stronger demonstration of the slot-filling mechanism than the calls-only analysis could provide: individuals do not reorganize their allocation curve around specific social partners, but instead maintain pre-existing relational slots and dynamically slot newly acquired alters into vacated functional positions.

Sixth, our structural panel models shed light on the main drivers of signature stability over time. Multilevel linear mixed-effects panel models show that alter turnover ($\beta = 0.349, p < 0.0001$) and percentage activity shifts ($p < 0.0001$) remain the primary drivers of temporal variation in signature self-divergence, with the turnover coefficient essentially unchanged from the calls-only estimate ($\beta = 0.351$). Local ego network clustering again shows no robust effect net of turnover, while baseline Negative Emotionality continues to predict greater signature stability ($\beta = -0.018, p = 0.012$). This indicates that signature stability is governed by individual-level relational habits rather than the triadic closure of the surrounding peer group.

Finally, personality modeling shows that social signature steepness remains anchored in Negative Emotionality, though the relationship is weaker once all channels are pooled. In parametric linear regressions, Negative Emotionality continues to predict steeper, core-concentrated allocation profiles ($\beta = 0.076, p = 0.0016$), but the overall model explains substantially less variance than under calls alone ($R^2 = 0.058$ versus 0.124), and the group-mean difference in top-rank allocation between high and low Negative Emotionality students that was visually striking under calls alone (31.7% versus 25.3% at Rank 1) is no longer apparent in the pooled data (34.3% versus 34.9%), indicating that the association is now diffuse across the tail of the distribution rather than concentrated at the primary alter. A classification and regression tree (CART) still roots on Negative Emotionality, but its cross-validated predictive performance is weak (relative error exceeds 1.0 at every candidate split), and the non-linear Extraversion-moderation pattern identified under calls alone—in which outgoing sociability buffered against hyper-concentration—is not robust to pooling text-based channels. We therefore treat the calls-only CART findings as channel-specific rather than as a general feature of personality-signature coupling, and report the pooled-data personality results with corresponding caution.

Turning to limitations and future work, while this study offers an empirical synthesis of social signature dynamics, several methodological and theoretical limitations suggest promising directions for future research. We address five essential analytical dimensions in sequence: operationalization and measurement constraints, threats to causal inference, directionality and selection feedback loops, temporal granularity and panel attrition, and institutional scope conditions.

Regarding operationalization and measurement constraints, our pooled communication measure treats a single outgoing call, SMS, MMS, or WhatsApp message as an equivalent unit of directed communicative effort. This equal-weighting assumption is a simplification: a multi-minute phone call and a one-word text plausibly reflect different magnitudes of relational investment, yet our count-based measure cannot distinguish them. Because text messages constitute 97.7% of all logged events (versus 2.3% for voice calls), the pooled signature is numerically dominated by texting behavior, and the compositional reversal we document at Rank 1 (from kinship-dominated under calls alone to friend-dominated once pooled) is at least partly a consequence of this high-volume, low-marginal-cost channel outweighing the comparatively rarer, higher-investment calls that anchor contact with parents and other long-duration kin. To evaluate this directly, Supplementary Table S1 recomputes the headline persistence, scaling, and compositional results separately for the voice-call

channel and the combined text-based channels. The core persistence finding is channel-invariant ($d_{\text{self}} < d_{\text{ref}}$, $p < 0.0001$ for calls alone, texts alone, and the pooled measure), but the compositional and curvature results are not: the call-only channel reproduces a kinship-dominated, power-law-preferred, more stable primary slot (72.1% family at Rank 1, 96.6% power-law preference, 59.9% top-alter retention) closely matching prior calls-only analyses of this cohort, whereas the text-only channel produces a friend-dominated, more volatile primary slot with a steeper decay exponent and weaker power-law preference (19.5% family, 56.1% friend, 38.2% power-law preference, 41.0% retention); because texts dominate the pooled measure by volume, the pooled results track the text-only channel closely on every metric. Prior multichannel studies similarly show that social signatures persist within voice and text channels evaluated separately, but that alter composition across channels exhibits limited overlap; the pooled measure used here cannot distinguish whether an individual tie is sustained by frequent brief texts, occasional substantive calls, or some mixture of the two. Future work should weight events by estimated relational investment (e.g., call duration, message length, or response latency) rather than treating all events as equivalent, and should report both single-channel and pooled results so that readers can assess the sensitivity of specific findings, such as the family-versus-friend composition of the primary slot, to this measurement choice.

Threats to causal inference and contextual confounding also warrant careful consideration. In residential university environments, social interactions are shaped by exogenous organizational structures, including freshman residence hall assignments, academic major tracks, dining hall layouts, and classroom schedules. While our study models individual-level dispositions and ego network metrics, unmeasured architectural and institutional opportunity structures could constrain or facilitate interaction opportunities independent of an individual’s intrinsic relational preferences. Future work should pair passive sensing with spatial positioning and administrative records to disentangle voluntary relationship maintenance from environmentally induced contact.

A related challenge concerns directionality, selection, and feedback loops. While our models establish that baseline Negative Emotionality predicts steeper signature concentration, the causal interplay between psychological states and communication structure may be bidirectional. An individual high in negative emotionality may actively retreat into an exclusive relationship to manage anxiety; conversely, becoming structurally dependent on an exclusive social tie may exacerbate negative emotionality and hinder broader socialization. Future research should implement continuous-time structural equation models or cross-lagged panel models to evaluate how fluctuations in signature steepness co-evolve with psychological distress and emotional states over time.

Issues of temporal granularity and panel attrition present additional analytical considerations. Our calendar windows follow institutional collegiate semesters, capturing the natural rhythms of undergraduate academic life. However, semester windows aggregate interaction dynamics over six-month periods, potentially smoothing out rapid, micro-temporal churn during intense examination periods, athletic seasons, or vacation breaks. Combining continuous passive logging with high-frequency ecological momentary assessment (EMA) could capture how social signatures adapt to short-term stressors. Furthermore, while the NetHealth study maintains high longitudinal

compliance, panel attrition over multi-year undergraduate studies requires ongoing attention to potential survival bias among highly compliant participants.

Finally, there is the institutional scope conditions of this study. Our analytic sample consists of undergraduate students at an elite, residential, private American university (the University of Notre Dame). Young adulthood represents a unique developmental phase characterized by geographic relocation, high relational mobility, identity formation, and intensive peer exploration. While this transitional context provides an ideal testing ground for signature stability under alter turnover, the empirical parameters observed here may not generalize directly to working adults, older populations, or non-residential community settings where relational turnover is substantially lower and family obligations are more institutionalized. Comparative cross-national and cross-institutional studies across diverse demographic strata will be essential to establish the global boundaries of social signature morphology.

Beyond these methodological considerations, our results carry significant theoretical implications for social network analysis, evolutionary anthropology, and computational network science: cognitive templates and capital. For decades, theories of social capital in social network analysis have emphasized the strategic advantages of network expansion, weak ties, and structural brokerage [37]. Under these classical formulations, social actors are portrayed as flexible investors who build, expand, and reshape their relational portfolios to maximize instrumental returns [38]. Conversely, evolutionary anthropology has emphasized biological and cognitive constraints, positing that human neocortical capacity strictly bounds the volume of relationships an individual can track [5, 7].

Our findings help bridge these perspectives by showing how structural flexibility and cognitive constraint coexist via persistent allocation templates. Individuals do not manage networks on an unconstrained, dyad-by-dyad basis; rather, they deploy enduring cognitive templates that allocate relational investment across structured role relations [39]. When social turnover occurs, individuals preserve their structural signature by seamlessly slotting new social entrants into pre-existing functional positions. The slot-filling dynamic reveals that personal networks possess a homeostatic structural equilibrium: the specific alters occupying the network are fluid and disposable, but the relational geometry governing their attention is rigid and enduring.

Furthermore, our integration of ego-network survey evaluations with continuous digital communication logs clarifies the micro-foundations of the social signatures phenomenon, linking it to classical issues of tie strength in social network analysis [37]. Following classical formulations [32, 33, 37], tie strength has long been conceptualized as a multidimensional construct encompassing emotional intensity, intimacy, reciprocal support, and longevity. Our findings show that social signatures reflect the behavioral enactment of this multidimensionality: an alter’s position slot in the signature is determined not merely by situational communication opportunities, but by the joint confluence of relationship longevity (accumulated biographical capital), cognitive recall salience (top-of-mind mental accessibility), and subjective intimacy. High-bandwidth communication serves as the ongoing behavioral mechanism that sustains this core, while lower-bandwidth peripheral ties provide flexible avenues for companionship and social exploration.

Finally, uncovering the psychological foundations of signature shape bridges the gap between individual personality dispositions [40] and network morphology, though our pooled-channel results counsel some humility about how far that bridge extends. Social signature steepness is not wholly disconnected from individual dispositions; the discovery that Negative Emotionality predicts core concentration, replicated (if attenuated) across both the calls-only and pooled multichannel measures, indicates that individuals prone to anxiety and emotional vulnerability adopt a somewhat more defensive, risk-averse allocation strategy, focusing relatively more of their scarce emotional resources on a trusted confidant. At the same time, the non-linear moderating role for Extraversion suggested by the calls-only decision tree did not survive pooling in text-based channels, and the overall share of variance in signature steepness explained by Big Five traits fell by more than half once texting was included. This suggests that psychological dispositions shape relational allocation more clearly, and perhaps more validly, within a single high-investment channel like voice calling than across the full, heterogeneous mix of mediated communication that constitutes most of young adults’ daily contact. In clinical and applied settings, tracking personal social signatures through passive digital sensing may still offer a promising, non-invasive method for identifying disruptions in relational equilibrium, but our results suggest that channel-specific signatures, rather than an undifferentiated pooled measure, may carry the more diagnostically useful psychological signal.

6 Methods

6.1 The NetHealth Cohort

The *NetHealth* study recruited an entire incoming cohort of first-year undergraduate students at the University of Notre Dame beginning in Fall 2015 [21–23]; comprehensive study design, instrumentation, and protocols are documented on the project website at <https://sites.nd.edu/nethealth/>. Participants consented to continuous digital sensing via their smartphones and Fitbit devices, alongside dense ego-network and psychological surveys administered at regular intervals throughout their college careers.

Our analytical design links five primary data streams. First, continuous communication logs capture pooled outgoing communication events collected from participant iOS devices spanning June 30, 2015, through July 2, 2017, across 515 unique egos. Second, comprehensive institutional academic calendar records document semester start and end dates, examination periods, orientation, and vacation breaks. Third, eight longitudinal waves of ego-network surveys record 35,913 alter nominations, capturing tie categories (family kin versus peer friends), perceived closeness, interpersonal trust, and multidimensional social support. Fourth, complete alter-alter edge lists document 174,748 structural ties connecting peers within each respondent’s personal egocentric network across survey waves. Finally, longitudinal psychometric survey batteries measure the Big Five personality dimensions (Extraversion, Agreeableness, Conscientiousness, Negative Emotionality [historically termed Neuroticism], Openness).

6.2 Pooling Communication Channels

A central methodological decision in this study concerns the scope of the communication channel analyzed. The underlying NetHealth communication-events database logs four categories of mediated interaction throughout the study period: voice calls, Short Message Service (SMS) texts, Multimedia Messaging Service (MMS) texts, and WhatsApp messages. Across the complete logged corpus ($N = 60,486,564$ events, all egos and event types), voice calls are a numerically minor channel, accounting for only 2.3% of all events (1,419,025 calls), whereas text-based channels overwhelmingly predominate: SMS messages alone constitute 87.2% of all logged events (52,719,968 messages), followed by WhatsApp messages (7.3%, 4,388,256) and MMS (3.2%, 1,959,315). Text-based channels combined therefore account for 97.7% of all communication events recorded in NetHealth, versus 2.3% for voice calls.

Because an exclusive focus on voice calls would characterize the structure of a numerically marginal channel rather than students' overall relational investment, we pool all four event types into a single measure of communication effort: each outgoing call, SMS, MMS, or WhatsApp message directed from ego i to alter j counts as one unit of interaction, regardless of channel. This treats a brief text and a sustained phone call as equivalent units of directed communicative effort, a simplifying assumption whose implications we revisit in the Discussion. Before pooling, we apply three data-quality exclusions recommended by the NetHealth data documentation: records with zero call duration, zero message length, or zero message size (which are retained when the duration/length/bytes field is missing rather than exactly zero, since missingness does not indicate the absence of content), and WhatsApp group-chat messages (which are not dyadic exchanges directed at a single alter). After restricting to outgoing events from iPhone devices (iPhones log all four channels consistently, whereas Android devices in NetHealth lack voice call and WhatsApp logs) within the June 30, 2015 through July 2, 2017 analytic window, the pooled sample comprises 15,609,345 communication events across 515 egos: 91.5% SMS, 3.3% WhatsApp, 2.7% voice calls, and 2.5% MMS.

6.3 Temporal Window Definitions

To establish whether social signatures persist across differing temporal resolutions, we construct two complementary binning schemes. The first scheme relies on calendar-based windows spanning July 1, 2015 through June 30, 2017. This includes two academic years (July 2015–June 2016 and July 2016–June 2017), four collegiate semesters (Fall 2015, Spring 2016, Fall 2016, and Spring 2017), eight academic quarters (July–September, October–December, January–March, and April–June), and 24 consecutive calendar months. The second scheme evaluates week-based windows spanning July 6, 2015 through July 2, 2017 across 104 Monday-to-Sunday weeks. This includes 102 three-week rolling windows evaluated in one-week increments, 52 discrete two-week intervals, and 104 discrete one-week intervals.

To ensure that estimated signatures reflect meaningful personal communication structures, egos are required to meet two standard inclusion criteria: (a) a minimum of 2 active alters in every

window ($k_{i,w} \geq 2$), and (b) an average pooled event volume exceeding 10 events per window. As summarized in Table 2, while the overarching sensing pool comprises $N = 515$ unique egos with outgoing communication events, the number of eligible participants naturally varies across timescales depending on the number of consecutive observation intervals required. The broadest calendar resolution (academic years, $W = 2$) retains $N = 405$ eligible egos. The collegiate semester timescale ($W = 4$) retains $N = 307$ participants, serving as our primary analytical sample for all subsequent dyadic, structural, and psychological modeling. Monthly and rolling weekly windows impose stricter longitudinal compliance across 24 to 104 intervals, retaining $N = 208$ and $N = 187$ egos, respectively. When directly benchmarking stability metrics across calendar resolutions, we evaluate the common calendar cohort ($N = 208$ egos) present in all four schemes simultaneously. Because pooling in high-volume text channels makes the > 10 events per window threshold easier to satisfy than under calls alone, eligible cohorts are uniformly larger than in a calls-only analysis, most dramatically for the narrowest window (1-week discrete: $N = 121$, compared with 14 under calls alone).

Table 2: NetHealth Cohort Summary Across Temporal Window Definitions

Window Scheme	Window Span	Windows (W)	Eligibility Criteria	Eligible Egos (N)
Academic Years	Jul 2015 – Jun 2017	2	≥ 2 alters, > 10 events	405
Semesters	Jul 2015 – Jun 2017	4	≥ 2 alters, > 10 events	307
Quarters	Jul 2015 – Jun 2017	8	≥ 2 alters, > 10 events	259
Months	Jul 2015 – Jun 2017	24	≥ 2 alters, > 10 events	208
Common Calendar Cohort	Jul 2015 – Jun 2017	–	All 4 calendar schemes	208
3-Week Rolling	Jul 2015 – Jul 2017	102	≥ 2 alters, > 10 events	187
2-Week Discrete	Jul 2015 – Jul 2017	52	≥ 2 alters, > 10 events	180
1-Week Discrete	Jul 2015 – Jul 2017	104	≥ 2 alters, > 10 events	121

6.4 Signatures and Divergence Metrics

For ego i in temporal window w , communication weights $w_{i,j,w}$ (measured as total pooled outgoing communication events: calls, SMS, MMS, and WhatsApp messages) to alters $j = 1, \dots, k_{i,w}$ are sorted in descending order:

$$w_{i,(1),w} \geq w_{i,(2),w} \geq \dots \geq w_{i,(k),w} \quad (9)$$

The social signature is defined as the normalized vector of interaction proportions:

$$p_{i,w}(r) = \frac{w_{i,(r),w}}{W_{i,w}}, \quad \text{where } W_{i,w} = \sum_{j=1}^{k_{i,w}} w_{i,j,w} \quad \text{and} \quad \sum_{r=1}^{k_{i,w}} p_{i,w}(r) = 1 \quad (10)$$

where $r \in \{1, 2, \dots, k_{i,w}\}$ represents the alter rank.

To quantify the similarity or difference between two probability vectors $P = \{p_r\}_{r=1}^{k_1}$ and $Q = \{q_r\}_{r=1}^{k_2}$ of potentially unequal lengths $k_1 \neq k_2$, we append zeros to the shorter distribution up to length $K = \max(k_1, k_2)$ on-the-fly and compute the *Jensen-Shannon Divergence* [JSD; 25]:

$$\text{JSD}(P \parallel Q) = H(M) - \frac{1}{2} [H(P) + H(Q)] \quad (11)$$

where $M = \frac{1}{2}(P + Q)$ is the mixture distribution, and $H(P)$ denotes Shannon entropy measured in bits:

$$H(P) = - \sum_{r=1}^K p_r \log_2(p_r) \quad (\text{with } 0 \log_2 0 \equiv 0) \quad (12)$$

JSD is symmetric ($\text{JSD}(P \parallel Q) = \text{JSD}(Q \parallel P)$), non-negative, and strictly bounded between 0 (identical distributions) and 1 bit (disjoint distributions).

For evaluating consistency across multiple observation windows simultaneously, we also compute the generalized n -ary Jensen-Shannon divergence across a set of n distributions $\{P_1, P_2, \dots, P_n\}$:

$$\text{JSD}(P_1, \dots, P_n) = H\left(\frac{1}{n} \sum_{m=1}^n P_m\right) - \frac{1}{n} \sum_{m=1}^n H(P_m) \quad (13)$$

We evaluate stability by defining two core divergence metrics. First, self-divergence (d_{self}) measures the mean pairwise JSD between consecutive temporal windows for the same ego across observation periods:

$$d_{\text{self}}(i) = \frac{1}{W-1} \sum_{w=1}^{W-1} \text{JSD}(P_{i,w} \parallel P_{i,w+1}). \quad (14)$$

Second, reference-divergence (d_{ref}) quantifies the divergence between the social signatures of distinct individuals observed within the same temporal window. This can be evaluated at the level of individual egos or across all unique participant pairs:

$$\text{By Ego: } d_{\text{ref}}(i) = \frac{1}{W} \sum_{w=1}^W \left[\frac{1}{n_w - 1} \sum_{j \neq i} \text{JSD}(P_{i,w} \parallel P_{j,w}) \right], \quad (15)$$

$$\text{All Pairs: } \bar{d}_{\text{ref}} = \frac{1}{\binom{n}{2}} \sum_{i < j} \left[\frac{1}{W} \sum_{w=1}^W \text{JSD}(P_{i,w} \parallel P_{j,w}) \right]. \quad (16)$$

If personal social signatures exhibit persistent idiosyncratic stability over time, an individual's self-divergence must be systematically and significantly smaller than the reference divergence observed between random peers ($d_{\text{self}} < d_{\text{ref}}$).

6.5 Ethics Declaration

The *NetHealth* study was reviewed and approved by the Institutional Review Board of the University of Notre Dame, and all data collection protocols (continuous smartphone sensing, Fitbit monitoring, and survey administration) were conducted in accordance with the approved protocol and relevant institutional guidelines, as documented at <https://sites.nd.edu/nethealth/>. All participants provided informed consent prior to enrollment. The present analysis uses de-identified secondary data released by the NetHealth research team under a case-coding system that anonymizes participant identities while preserving the ability to link records across data streams; no additional identifiable information was accessed by the authors.

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Author Contributions

O.L. and D.H. conceived the study. D.H. directed the NetHealth data collection infrastructure. O.L. designed the analytic pipeline, conducted the statistical analyses, and drafted the manuscript. Both authors interpreted the results, revised the manuscript, and approved the final version.

Data Availability Statement

The NetHealth communication-events, ego-network survey, and basic survey data analyzed in this study are available from the NetHealth project upon completion of a data-use registration at <https://sites.nd.edu/nethealth/data-2/>. Processed derivative datasets, analysis code, and the full per-channel robustness supplement underlying this manuscript are available from the corresponding author upon reasonable request.

Competing Interests

The authors declare no competing interests.